Continuity

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Testing continuity**

Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.

a Find $\lim_{x \to 2^-} f(x)$. $= 5$.

b Find $\lim_{x \to 2^+} f(x)$. $= 5$.

c Is f continuous at $x = 2$? Justify. Yes — both one-sided limits equal $5 = f(2)$, so f is continuous at $x = 2$.

(a) $= 5$.; (b) $= 5$.; (c) Yes — both one-sided limits equal $5 = f(2)$, so f is continuous at $x = 2$.

2 **[Foundational]** • **Target: Testing continuity**

The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k.

Need $3k + 1 = 9 - 2 = 7$, so $k = 2$.

3 **[Foundational]** • **Target: Types of discontinuity and the Intermediate Value Theorem**

Classify the discontinuity of each function at the given point:

a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$ Removable (a hole) — the factor $(x - 1)$ cancels.

b $f(x) = \frac{1}{x - 2}$ at $x = 2$ Infinite discontinuity — f blows up as $x \to 2$.

c $f(x) = \lfloor x \rfloor$ at $x = 3$ Jump discontinuity — the floor function jumps from 2 to 3.

(a) Removable (a hole) — the factor $(x - 1)$ cancels.; (b) Infinite discontinuity — f blows up as $x \to 2$.; (c) Jump discontinuity — the floor function jumps from 2 to 3.

4 **[Foundational]** • **Target: Types of discontinuity and the Intermediate Value Theorem**

Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.

$f(1) = 1 - 4 + 1 = -2 < 0$ and $f(2) = 8 - 8 + 1 = 1 > 0$. Since f is continuous (a polynomial) and changes sign, the IVT guarantees a root in $(1, 2)$.

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5 [Foundational] • Target: Continuity on an interval

State the largest interval(s) on which each function is continuous:

a $f(x) = \frac{1}{x-4}$ Continuous everywhere except $x = 4$: on $(-\infty, 4) \cup (4, \infty)$.

b $f(x) = \sqrt{x-3}$ Continuous on $[3, \infty)$.

c $f(x) = \ln(x+2)$ Continuous on $(-2, \infty)$.

(a) Continuous everywhere except $x = 4$: on $(-\infty, 4) \cup (4, \infty)$.; (b) Continuous on $[3, \infty)$.; (c) Continuous on $(-2, \infty)$.

6 [Foundational] • Target: Continuity on an interval

Find the value of c that makes $f(x) = \frac{x^2-9}{x-3}$ for $x \neq 3$, and $f(3) = c$, continuous at $x = 3$.

$\lim_{x \to 3} \frac{x^2-9}{x-3} = \lim_{x \to 3} (x+3) = 6$, so $c = 6$.

7 [Foundational] • Target: Applying the Intermediate Value Theorem

Show that $f(x) = \cos x - x$ has a root in the interval $(0, 1)$.

$f(0) = \cos 0 - 0 = 1 > 0$ and $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$. Since f is continuous and changes sign on $[0, 1]$, IVT guarantees a root in $(0, 1)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Applying the Intermediate Value Theorem

A hiker starts a trail at elevation 200 m at 8:00 AM and finishes at elevation 800 m at noon. Explain, using the IVT, why the hiker must have been at elevation exactly 500 m at some point during the hike.

Elevation as a function of time is continuous (the hiker moves without teleporting), starts below 500 m and ends above 500 m, so by the IVT the hiker's elevation must equal 500 m at some intermediate time.

9 [Intermediate] • Target: Visualizing discontinuities

The function $f(x) = \frac{1}{x-2}$ (with an infinite discontinuity at $x = 2$) is graphed below. Use the graph to describe the one-sided behavior near $x = 2$.

As $x \to 2^-$, $f(x) \to -\infty$; as $x \to 2^+$, $f(x) \to +\infty$ — the function is unbounded on both sides near $x = 2$, confirming an infinite (non-removable) discontinuity.

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10 [Intermediate] • Target: Continuity of composite and combined functions

Explain why the composition of two continuous functions is continuous, and use this to argue that $h(x) = \sqrt{x^2 + 1}$ is continuous everywhere.

If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a (limits pass through continuous functions). Here $x^2 + 1$ is a continuous polynomial (always $\geq 1 > 0$), and $\sqrt{u}$ is continuous for $u \geq 0$, so the composition $\sqrt{x^2 + 1}$ is continuous for all real x.

11 [Intermediate] • Target: Removable discontinuities and redefinition

The function $f(x) = \frac{x^2 + x - 6}{x - 2}$ has a removable discontinuity at $x = 2$. Find the value that, if assigned to $f(2)$, would make f continuous there.

Simplify: $\frac{(x-2)(x+3)}{x-2} = x + 3$ for $x \neq 2$. $\lim_{x \to 2} f(x) = 2 + 3 = 5$. Assigning $f(2) = 5$ removes the discontinuity.

12 [Intermediate] • Target: Removable discontinuities and redefinition

Find all values of x where $f(x) = \frac{x^2 - 16}{x^2 - 2x - 8}$ has a discontinuity, and classify each as removable or non-removable.

Factor: $\frac{(x-4)(x+4)}{(x-4)(x+2)}$. Common factor $(x - 4)$: removable discontinuity at $x = 4$. Remaining denominator zero at $x = -2$: non-removable (infinite) discontinuity.

13 [Intermediate] • Target: The Extreme Value Theorem (a continuity consequence)

State the Extreme Value Theorem, and explain why $f(x) = x^2$ is guaranteed to have an absolute maximum and minimum on $[-3, 2]$.

The EVT states a continuous function on a closed interval $[a, b]$ attains both an absolute max and min on that interval. Since $f(x) = x^2$ is continuous (a polynomial) and $[-3, 2]$ is closed, the EVT guarantees both extrema exist (here: min at $x = 0$, max at $x = -3$).

14 [Intermediate] • Target: Continuity of composite and combined functions

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

Set up Lagrange system $ablaf = \lambda ablag$: $ablaf = \langle 2x, 4y \rangle$, $ablag = \langle 2x, 2y \rangle$. 1) $2x = 2\lambda x \Rightarrow 2x(1 - \lambda) = 0$ 2) $4y = 2\lambda y \Rightarrow 2y(2 - \lambda) = 0$ 3) $x^2 + y^2 = 1$ From (1), either $x = 0$ or $\lambda = 1$: • If $x = 0$: from constraint (3), $y^2 = 1 \Rightarrow y = \pm 1$. Points: $(0, 1)$ and $(0, -1)$. Value: $f(0, \pm 1) = 0 + 2(1)^2 = 2$. • If $\lambda = 1$: from (2), $2y(2 - 1) = 2y = 0 \Rightarrow y = 0$. From constraint (3).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

Level 2 · Medium Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the counterclockwise circle $x^2 + y^2 = 4$.

Identify components: $P = 3y$ and $Q = 5x$. Compute partial derivatives: $\frac{\partial Q}{\partial x} = 5$ and $\frac{\partial P}{\partial y} = 3$. Apply Green's Theorem: $\oint_C (3y\,dx + 5x\,dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \iint_R (5 - 3)\,dA = 2\iint_R dA$. The region R is a disk of radius $r = 2$, so its area is $\pi(2^2) = 4\pi$. Therefore, $2\iint_R dA = 2(4\pi) = 8\pi$.

16 **[Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 11.01 Sequences and Series**

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ for the function: $y = \int_{\cos x}^{x^3} \sqrt{1 - t^2}\,dt$.

Apply the Leibniz Rule with upper limit $u(x) = x^3 \Rightarrow u'(x) = 3x^2$ and lower limit $v(x) = \cos x \Rightarrow v'(x) = -\sin x$: $\frac{dy}{dx} = \sqrt{1 - (x^3)^2} \cdot (3x^2) - \sqrt{1 - (\cos x)^2} \cdot (-\sin x)$. Simplify trigonometric terms with $\sqrt{1 - \cos^2 x} = \sqrt{\sin^2 x} = |\sin x|$: $\frac{dy}{dx} = \mathbf{3x^2\sqrt{1 - x^6} + |\sin x|\sin x}$.

17 **[Challenging] • Target: Testing continuity • Cross-Lesson: 12.01 Convergence Tests**

Level 1 · FoundationsFind $h'(2)$ given the accumulation function $h(x) = \int_0^{x^2} \frac{t}{t+1}\,dt$.

By FTC Part 1 and the Chain Rule, let $u = x^2 \Rightarrow \frac{du}{dx} = 2x$: $h'(x) = \frac{u}{u+1} \cdot \frac{du}{dx} = \frac{x^2}{x^2+1} \cdot (2x) = \frac{2x^3}{x^2+1}$. Evaluate at $x = 2$: $h'(2) = \frac{2(2^3)}{2^2+1} = \frac{2(8)}{4+1} = \mathbf{\frac{16}{5}}$.

18 **[Challenging] • Target: Types of discontinuity and the Intermediate Value Theorem • Cross-Lesson: 12.02 Power Series and Taylor Series**

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 **[Challenging] • Target: Continuity on an interval • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions**

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 **[Challenging] • Target: Applying the Intermediate Value Theorem • Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

Level 3 · AdvancedLet $f(x) = x^3 - 3x - 1$. Explain how the Intermediate Value Theorem guarantees a root in $[1, 2]$.

1. $f(x)$ is a polynomial, so it is continuous on $[1, 2]$. 2. $f(1) = 1 - 3 - 1 = -30$. 3. Since 0 lies strictly between $f(1) = -3$ and $f(2) = 1$, by the IVT there must exist at least one $c \in (1, 2)$ such that $f(c) = 0$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Testing continuity

A continuous function on a closed, bounded interval $[a, b]$ is guaranteed (Extreme Value Theorem) to be:

a. bounded and to attain its max and min ✓

b. monotonic

c. unbounded

d. differentiable

The Extreme Value Theorem: continuity on a compact interval guarantees both boundedness and attained extrema.

22 [Foundational] • Target: Types of discontinuity and the Intermediate Value Theorem

Removable discontinuity: $f(x) = \frac{x^2-4}{x-2}$. To make f continuous at $x = 2$, define $f(2) =$

a. 4 ✓

b. 2

c. 0

d. -4

Applying Limits and Continuity principles, $(x^2 - 4)/(x - 2) = x + 2 \to 4$. Therefore, the verified correct option is 4.

23 [Foundational] • Target: Continuity on an interval

By the Intermediate Value Theorem, does $f(x) = x^3 - x - 1$ have a root in $[1, 2]$?

a. No $--$ $f(1)$ and $f(2)$ have the same sign

b. No $--$ f is not continuous on $[1, 2]$

c. Yes $--$ $f(1)$ and $f(2)$ have opposite signs ✓

d. Cannot be determined without more information

$f(1) \approx -1.00$ and $f(2) \approx 5.00$ have opposite signs, and f is continuous, so IVT guarantees a root in $(1, 2)$.

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24 [Foundational] • Target: Applying the Intermediate Value Theorem

Does $f(x) = \frac{x-1}{x-1}$ have a removable discontinuity at $x = 1$?

a. No $--$ it is an infinite discontinuity

b. No $--$ f is continuous everywhere

c. No $--$ it is a jump discontinuity

d. Yes $--$ the factor cancels, leaving a hole ✓

$\frac{x-1}{x-1} = 1$ for $xe1$, but $f(1)$ is undefined -- a removable hole.

25 [Foundational] • Target: Visualizing discontinuities

Intermediate Value Theorem guarantees a root of $f(x) = x^3 - x - 2$ on:

a. $(-1, 0)$

b. $(0, 1)$

c. $(1, 2)$ ✓

d. $(2, 3)$

$f(1) = -20$; IVT guarantees $f(c) = 0$ for some $c \in (1, 2)$.

26 [Foundational] • Target: Continuity of composite and combined functions

Global extrema on closed interval: Extreme Value Theorem guarantees max and min are attained. Find them by checking:

a. **all critical points and endpoints** ✓

b. only critical points

c. only endpoints

d. only inflection points

EVT: critical points + endpoints.

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27 [Foundational] • Target: Removable discontinuities and redefinition

If f has a jump discontinuity at $c \in (a, b)$, the accumulation function $F(x) = \int_a^x f(t)dt$ is still:

a. continuous at c (though not differentiable there) ✓

b. discontinuous at c

c. differentiable at c

d. undefined at c

Integrating a bounded (even discontinuous) function always yields a continuous accumulation function; differentiability can fail at the jump.

28 [Foundational] • Target: The Extreme Value Theorem (a continuity consequence)

Extreme Value Theorem: continuous f on $[a, b]$ attains:

a. only a maximum

b. only a minimum

c. **both max and min** ✓

d. neither

Applying Applications of Derivatives principles, EVT guarantees both. Therefore, the verified correct option is both max and min.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Testing continuity

If F and G are both antiderivatives of the same continuous f on an interval, then $F - G$ is:

a. **a constant function** ✓

b. strictly increasing

c. equal to f

d. not necessarily differentiable

$(F - G)' = f - f = 0$ everywhere, so by the MVT, $F - G$ is constant — this is why FTC Part 2's antiderivative is unique only up to $+C$.

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30 [Intermediate] • Target: Types of discontinuity and the Intermediate Value Theorem

If f is continuous on $[0, 1]$ with $f(x) \in [0, 1]$ for all x, the Brouwer Fixed-Point Theorem guarantees:

a. f is differentiable

b. $f(x) = 0$ for some x

c. $f(c) = c$ **for some** $c \in [0, 1]$ ✓

d. f is monotone

Consider $g(x) = f(x) - x$: $g(0) = f(0) \geq 0$ and $g(1) = f(1) - 1 \leq 0$. IVT gives $g(c) = 0$, i.e., $f(c) = c$.

31 [Intermediate] • Target: Continuity on an interval

Find the hole (removable discontinuity) of $f(x) = \frac{x^2 + x - 2}{x^2 + 2x - 3}$.

a. x=-3

b. x=-2

c. **x=1** ✓

d. no hole

Factor: $\frac{(x-1)(x+2)}{(x-1)(x+3)}$; the common factor $(x - 1)$ cancels, giving a hole at $x = 1$.

32 [Intermediate] • Target: Applying the Intermediate Value Theorem

Classify the discontinuity of $f(x) = \frac{x^2 - 9}{x - 3}$ at $x = 3$.

a. jump discontinuity

b. infinite discontinuity

c. no discontinuity

d. **removable discontinuity** ✓

The numerator shares a factor $(x - 3)$ with the denominator, which cancels -- removable.

33 [Intermediate] • Target: Visualizing discontinuities

A step function like the greatest-integer function $f(x) = \lfloor x \rfloor$ has ___ discontinuities at every integer.

a. infinite

b. removable

c. no

d. **jump** ✓

At each integer, the left and right limits differ by exactly 1 -- a jump discontinuity.

34 [Intermediate] • Target: Continuity of composite and combined functions

For the Extreme Value Theorem to guarantee a global max and min, f must be continuous on:

a. all real numbers

b. an open interval

c. a single point

d. **a closed interval** ✓

The Extreme Value Theorem requires continuity on a closed, bounded interval $[a, b]$.

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35 [Intermediate] • Target: Removable discontinuities and redefinition

A function with a vertical asymptote at $x = 2$ has a(n) ___ discontinuity there.

a. removable

b. jump

c. **infinite ✓**

d. oscillatory

Vertical asymptotes create infinite discontinuities where $f \to \pm\infty$.

36 [Intermediate] • Target: The Extreme Value Theorem (a continuity consequence)

If $f(x) = \frac{x^2 - 9}{x - 3}$, which type of discontinuity occurs at $x = 3$?

a. Jump

b. Infinite

c. **Removable ✓**

d. Essential

The factor $(x - 3)$ cancels; the hole is removable by defining $f(3) = 6$.

37 [Intermediate] • Target: Testing continuity

If f has a jump discontinuity at $c \in (a, b)$, then $G = \int_a^x f\,dt$ at c is:

a. discontinuous

b. differentiable

c. **not differentiable but continuous ✓**

d. zero

G remains continuous but loses differentiability at the jump.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Types of discontinuity and the Intermediate Value Theorem • Cross-Lesson: 12.02 Power Series and Taylor Series

Which function has a jump discontinuity at $x = 0$?

a. $f(x) = x^2$

b. $f(x) = 1/x$

c. $f(x) = |x|/x$ ✓

d. $f(x) = \sin x$

$|x|/x = \pm 1$ gives $\lim^{+} = 1$ and $\lim^{-} = -1$, so it jumps.

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39 [Challenging] • Target: Continuity on an interval • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Change of variables $\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$ is a consequence of:

a. MVT

b. **chain rule + FTC ✓**

c. IBP

d. Cauchy's integral formula

$\frac{d}{dx}F(g(x)) = f(g(x))g'(x)$, integrate via FTC.

40 [Challenging] • Target: Applying the Intermediate Value Theorem • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Mean Value Theorem guarantees c in open interval — this means the point is:

a. an endpoint

b. **strictly interior ✓**

c. a maximum

d. undefined

Open interval (a, b): strictly interior.

41 [Challenging] • Target: Visualizing discontinuities • Cross-Lesson: 15.01 Double Integrals

Let f be continuous on $\mathbb{R}$ and $f(0) = 0$. If $f(x + y) = f(x)f(y)$ for all x, y, then:

a. $f(x) = 0$ everywhere

b. $f(x) = e^{cx}$ **for some** c **✓**

c. $f(x) = x$

d. $f(x)$ is periodic

Functional equation forces $f(x) = a^x = e^{x \ln a}$. With $f(0) = 0$ contradiction unless $f \equiv 0$. Actually $f(0) = f(0 + 0) = f(0)^2 \Rightarrow f(0) = 0$ or 1. If $f(0) = 0$ then $f(x) = f(x + 0) = f(x)f(0) = 0$. So $f \equiv 0$ or $f(x) = e^{cx}$.

42 [Challenging] • Target: Continuity of composite and combined functions • Cross-Lesson: 16.01 Vector Fields and Line Integrals

IBP formula follows from FTC because $\int_a^b fg'\,dx = [fg]_a^b - \int_a^b f'g\,dx$ derives from:

a. **Differentiating** fg **and applying FTC Part 2 ✓**

b. L'Hôpital

c. Green's theorem

d. Cauchy-Schwarz

Applying Fundamental Theorem principles, Product rule + FTC Part 2. Therefore, the verified correct option is Differentiating fg and applying FTC Part 2.

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43 [Challenging] • Target: Removable discontinuities and redefinition • Cross-Lesson: 2.01 Evaluating Limits

$\int_0^1 \frac{1}{x^p}\,dx$ (improper at 0) converges if and only if:

a. $p < 1$ ✓

b. $p > 1$

c. $p \le 0$

d. never converges

By FTC as a one-sided limit, $\left[\frac{x^{1-p}}{1-p}\right]_0^1$ is finite only when $1 - p > 0$, i.e. $p < 1$ — the mirror image of the p-test at infinity.

44 [Challenging] • Target: The Extreme Value Theorem (a continuity consequence) • Cross-Lesson: 3.01 The Derivative from First Principles

Newton's law of cooling: $T(t) = T_e + (T_0 - T_e)e^{-kt}$. If room temp $T_e = 20$, initial $T_0 = 80$, $k = 0.1$, $T(10) =$

a. $20 + 60$

b. $60e^{-1}$

c. $80e^{-1}$

d. $20 + 60e^{-1} \approx 42.1$ ✓

Applying Transcendental Functions principles, $T(10) = 20 + 60e^{-1} \approx 42.1°\text{C}$. Therefore, the verified correct option is $20 + 60e^{-1} \approx 42.1$.

45 [Challenging] • Target: Testing continuity • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Let $F(x) = \int_0^x (t^2 + 5t + 4)\,dt$. At which x does F have a local maximum?

a. -1

b. $-\frac{5}{2}$

c. 0

d. -4 ✓

$F'(x) = (x + 1)(x + 4)$. F' changes sign at -4 (from + to –) and at -1 (from – to +), so the local max is at $x = -4$.

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